Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 21)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Table 1 shows some events in the history of life on Earth.

Table 1

Event	Time before present / billion years
Formation of the Earth	4.6
First life (Archaea) on Earth	3.5
First photosynthetic organisms	2.7
Oxygen concentration in the atmosphere starts to rise	2.45
End of the period of oxygen build-up (Proterozoic)	0.54

0 1 . 1 Use Table 1 to calculate how many **million** years passed between the appearance of photosynthetic organisms and the start of the rise in atmospheric oxygen.

[1 mark]

Answer _____ million years

0 1 . 2 Explain why atmospheric oxygen did not start to rise as soon as photosynthesis began.

[2 marks]

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0 1 . 3 Explain how the rise in atmospheric oxygen made life on the land surface possible.

[3 marks]

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0 1 . 4 Explain how early photosynthetic organisms helped to prevent a long-term rise in the Earth's temperature.

[2 marks]

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0 1 . 5 Give **two** other ways in which living organisms changed the conditions on Earth.

[2 marks]

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2 .....

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0 2 . 1 Which of the following is a feature of natural systems? Tick **one** box.

[1 mark]

A Reactions that require high temperatures ☐

B Reliance on non-renewable energy ☐

C Waste products that are the raw materials for other processes ☐

D Dispersal of elements so they cannot be re-used ☐

0 2 . 2 Explain why nitrogen fixation by bacteria is more sustainable than the Haber process.

[2 marks]

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0 2 . 3 Explain why incineration of wastes is less sustainable than natural decomposition.

[2 marks]

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0 3 . 1 Explain why a moderate water shortage reduces crop productivity even though the plants survive.

[3 marks]

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0 3 . 2 Explain the problems caused by waterlogged soils and describe **two** ways farmers reduce soil water levels.

[4 marks]

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0 3 . 3 Explain how irrigating with groundwater can cause soil salinisation and how this harms crops.

[2 marks]

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0 3 . 4 State how a farmer could raise the pH of an acidic soil and explain why a low pH reduces fertility.

[1 mark]

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0 4 A conservation team is planning to protect a population of an endangered antelope in a reserve.

0 4 . 1 Explain why each of the following features of the population should be measured before a conservation strategy is decided.

[8 marks]

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Population size .....

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Distribution .....

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Survival rate .....

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Age structure .....

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0 4 . 2 Describe how the population size of the antelope could be estimated.

[2 marks]

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0 5 . 1 Describe **four** ways in which the Wildlife and Countryside Act (1981) protects wildlife in the UK.

[4 marks]

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0 5 . 2 Explain the difference between CITES Appendix I and Appendix II, giving an example species for each.

[4 marks]

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0 5 . 3 Explain why controlling international trade can be more effective than trying to stop collection of a species in its own country.

[2 marks]

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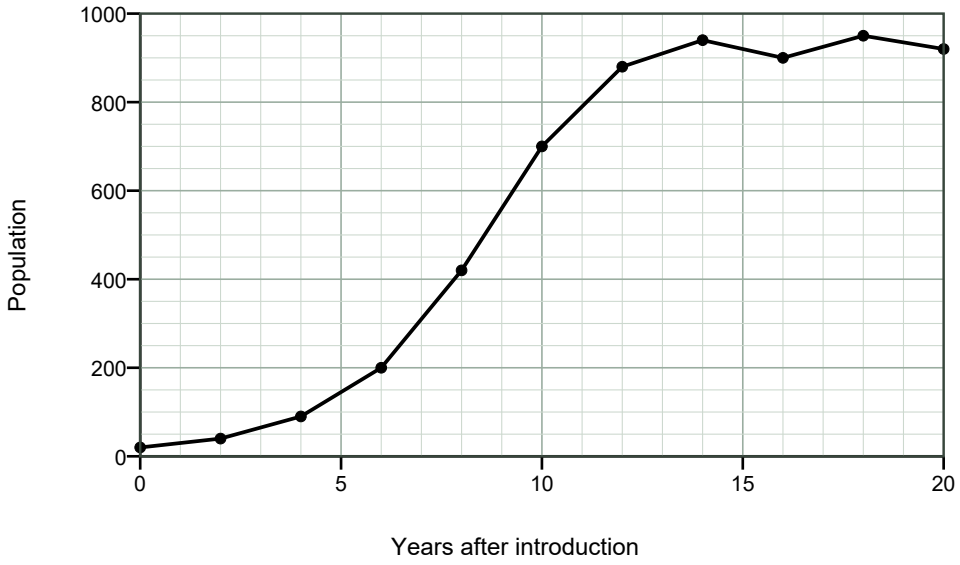
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0 6 Figure 1 shows the population of red deer on an island after they were introduced in year 0. No predators are present.

Figure 1



0 6 . 1 Name the shape of the growth curve and explain the changes in growth rate shown.

[4 marks]

0 6 . 2 Explain the difference between density-dependent and density-independent factors, with an example of each.

[2 marks]

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0 6 . 3 Explain why the deer population on the island may need to be culled even though it appears stable.

[2 marks]

0 6 . 4 Explain how predator and prey populations regulate each other.

[2 marks]

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0 7 A conservation organisation is converting 700 ha of arable farmland on the coast into a wetland reserve for breeding and wintering birds (as at Wallasea Island).

0 7 . 1 Explain why habitat area and shape are important in the design of the reserve.

[4 marks]

0 7 . 2 Explain how the reserve could be designed to provide a range of niches.

[3 marks]

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0 7 . 3 Explain why some species may need to be deliberately introduced while others colonise naturally.

[2 marks]

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0 7 . 4 Explain the value of a biological corridor linking the reserve to other wetlands.

[1 mark]

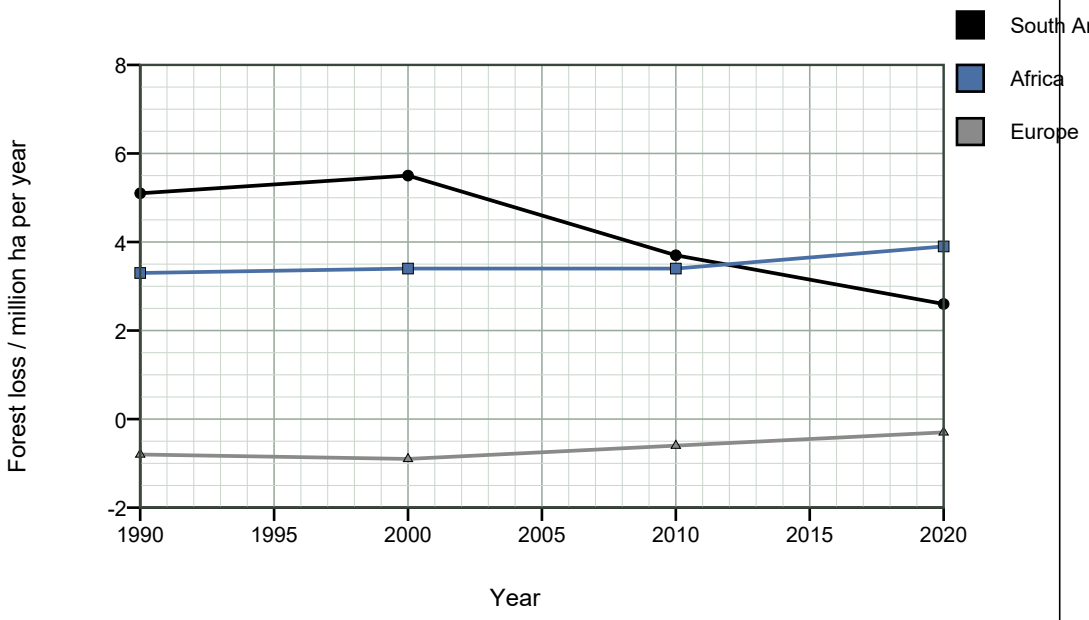
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0 8 Figure 1 shows the area of forest lost each year in three regions between 1990 and 2020.

Figure 1



0 8 . 1 Describe the trends shown in Figure 1.

[3 marks]

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0 8 . 2 Explain the main causes of forest loss in South America and Africa.

[4 marks]

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0 8 . 3 Explain why roads and railways cause more forest loss than the area they occupy.

[2 marks]

0 8 . 4 Explain why softwood from Scandinavia and Canada is generally produced more sustainably than tropical hardwood.

[1 mark]

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0 9 . 1 Explain how GPS mapping, drone surveys and computer-based yield monitoring can reduce the environmental impacts of agriculture.

[3 marks]

0 9 . 2 Explain why productive agricultural technologies are less available in LEDCs.

[2 marks]

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0 9 . 3 Explain how buying fairtrade food affects producers.

[1 mark]

0 9 . 4 Explain how nitrate from farmland can threaten human health.

[2 marks]

0 9 . 5 Give **two** agricultural sources of methane.

[2 marks]

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1 0 . 1 Evaluate the extent to which aquaculture can replace fishing as a source of aquatic food.

[9 marks]

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Extra space

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1 0 . 2 State what is meant by a **fed** aquaculture species.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably.

[25 marks]

OR

1 1 . 2 Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS